

Program : Diploma in Polymer Technology	
Course Code : 4098	Course Title: Latex Technology Lab
Semester : 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0,T:0,P:3)	Periods per semester: 45

Course Objectives:

- To provide expertise on quality assurance of NR latex as raw material for industrial applications.
- To provide hands-on experience for the students in the latex products manufacturing.
- To develop skills on different aspects of latex product manufacturing from compound preparation to after treatments which will be beneficial in future to work with industry or to start. entrepreneurial ventures.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics 1	1
		Mathematics II	2
Concentration of Natural Rubber Latex		Natural Rubber Production	3

Course Outcomes:

On completion of course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Practice various specification tests of Natural Rubber Latex as per IS:3708	20	Applying
CO2	Conduct concentration of field latex by creaming process.	3	Applying
CO3	Prepare dispersions and emulsions of various ingredients for latex compounding.	6	Applying

CO4	Perform manufacture of various latex products by dipping, foaming and casting	14	Applying
	Lab Exam	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3					
CO2	3						
CO3	3						
CO4	3	3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Practice various specification tests of Natural Rubber Latex as per IS:3708		
M1.01	Determination of DRC of NR latex	3	Applying
M1.02	Determination of TSC and NRC of NR latex	3	Applying
M1.03	Determination of Alkalinity of NR latex	3	Applying
M1.04	Determination of Magnesium content of NR latex	3	Applying
M1.05	Determination of coagulum content of NR latex	3	Applying
M1.06	Determination of sludge content of NR latex	2	Applying
M1.07	Determination of Specific gravity of NR latex	2	Applying
M1.08	Determination of viscosity of NR latex.	1	Applying
	Lab Exam I	1	
CO2	Conduct concentration of field latex by creaming process		
M2.01	Prepare 2% solution of creaming agent(tamarind seed powder, sodium/ammonium alginate)	1	Applying

M2.02	Carry out the creaming of NR Latex and check the Creaming level after 24 and 48 Hours	1	Applying
	Lab Exam I	1	
CO3	Prepare dispersions and emulsions of various ingredients for latex compounding		
M3.01	Prepare the dispersion of solid compounding ingredients for latex compounding.	3	Applying
M3.02	Prepare the emulsions of liquid compounding ingredients such as SP and process oil for latex products	3	Applying
CO4	Perform manufacture of various latex products by dipping, foaming and casting		
M4.01	Prepare latex compounds for various latex products	3	Applying
M4.02	Prepare thin walled products by latex dipping (finger cap, balloon etc)	3	Applying
M4.03	Prepare house hold gloves products by dipping	3	Applying
M4.04	Manufacture simple toys by casting process	3	Applying
M4.05	Produce foamed latex products	2	Applying
	Lab exam II	1	

Text / Reference:

T/R	Book Title/Author
T1	High Polymer Lattices, vol-I, vol-II, Vol-III - D.C.Blackley (Publisher – Springer Netherlands,1997)
R2	Latex in Industry - R.J.Noble (Publisher – Rubber Age, Palmerton, New York).
R3	Training Manual - RTI, RRIL, Kottayam
R4	Rubbery Materials and their compounds - J.A Brydson (Publisher –Elsevier) Applied Science, London, 1988)

Online Resources:

Sl. No	Website Link
1	https://www.jacionline.org
2	http://www.nocil.com
3	http://www.rubberscience.in .
4	https://www.researchgate.net
5	https://link.springer.com